

The Triple Demotion of the Nuclear Shell: The Shell Model, the Liquid Drop Model, and the Independent-Particle-Motion Axiom

——Derivational Demotion Based on 2D Ising Exact Solution Calibration, Continuous Sphericity Weight Recovery, and the QCTI Index

Shuiping Tang
(Independent Researcher)

Abstract

The central axiom of the traditional shell model—nucleons moving independently in a static mean field—and the liquid drop model have coexisted for nearly nine decades without resolution. Drawing on Tang (2026v) and Tang (2026w), this paper integrates five lines of evidence to complete a unified argument for the triple demotion of the nuclear shell: the shell model, the liquid drop model, and the independent-particle-motion axiom are projections of a quantum-classical hybrid in specific limits. The argument proceeds in five tiers.

Tier 1—2D Ising calibration: With T_c as the fixed breakpoint, the intercept/slope ratio across five temperature intervals decreases monotonically from 875.75 in the deep ferromagnetic region to 2.67 near T_c (Spearman $\rho = 1.0000$), establishing it as a metric of distance from the phase-transition critical point. The ratio is strictly negatively correlated with susceptibility proxy $|dM/dT|$ ($\rho = -1.0000$) and mean-field error ϵ_{MF} ($\rho = -1.0000$). The 2D Ising calibration heuristically supports the distance interpretation; the quantum-classical transition interpretation in nuclear physics is independently anchored by the strong negative correlation with shell-correction energy ΔR^2 ($\rho = -0.87$, $p = 0.0000$)—the two lines are independent yet convergent.

Tier 2—Quantum-classical hybrid: The Quantum-Classical Transition Index (QCTI) decreases monotonically from +0.94 in the light-nucleus region to -0.81 in the heavy-nucleus region ($\rho = -0.95$); a simplified QCTI using only shell-correction energy ΔR^2 and the binary marker yields $\rho = -0.93$. The seven-metric evidence matrix displays zonal characteristics across three nuclear regions. The double-track intersection $N \approx 70.8$ (spectrum value = 0.407) divides the N -axis into quantum-dominated and classical-dominated regions. The mid-shell region ($N=50 - 100$) is the common silence zone of all activation operators—Chow F drops to 6.8, only 1/15 of the light-nucleus region's 105.4 (Bootstrap 95% CI fully separated), with Kendall's W confirming complete agreement among the six operators. The mid-shell silence zone and the liquid-drop masking zone are negatively correlated ($\rho = -0.67$, $p = 0.0001$). The binary magic-number marker's local interaction effect decays significantly across 28 sliding windows ($\rho = -0.57$, $p = 0.0016$, a 132-fold decay), while shell-correction energy ΔR^2 peaks in the mid-shell region.

Tier 3—First demotion: the shell model. A continuous sphericity weight function based on distances to the nearest magic numbers is constructed. Weighted least squares on the nuclide data shows that the Chow F of the five traditional magic numbers is amplified by 3.68 – 9.68 times ($\sigma = 3.0$); amplification exceeds 1.5 for all five magic numbers for $\sigma = 2.0 - 5.0$ and decreases monotonically with increasing σ , demonstrating that the shell model is the projection of the regime-switch network in the spherical-symmetry limit.

Tier 4—Second demotion: the liquid drop model. The QCTI tends to -0.81 in the heavy-nucleus region, the monotonic relationship between the intercept/slope ratio and ϵ_{MF} ($\rho = -1.0000$) holds, and the uniformity of the liquid-drop masking effect supports the absorption hypothesis—together proving that the liquid drop model is the projection of the regime-switch network in the classical limit.

Tier 5—Third demotion: the independent-particle-motion axiom. Five lines of evidence from four independent dimensions establish the incompleteness of the axiom as a statistical description framework for many-body systems. The shell correction is re-positioned as a quantum residual—compensating for the difference between quantum shell effects and classical liquid-drop behavior, peaking in the mid-shell region.

This paper refines six theorems (Theorems A – F), with all 8 conditions for derivational demotion satisfied. All current evidence derives from multi-dimensional re-analyses of the same AME2020 nuclear mass data—statistically independent but not data-source independent; cross-data-source verification is a key direction for future work. This paper completes the derivational demotion in nuclear physics, which, together with the demotion of gauge field theory in particle physics and the demotion of the periodic law in chemistry, constitutes the complete demotion argument of the Factor Hierarchy Law in the three hard-science disciplines.

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1. Introduction

1.1 The Shell Model and the Liquid Drop Model—Nearly Nine Decades of Conceptual Tension

The nuclear shell model is the cornerstone of nuclear physics. Its central axiom is that nucleons move independently in a static mean field (Mayer & Jensen, 1955). A direct corollary of this axiom is the existence of magic numbers—when large energy gaps appear between single-particle levels, nuclei exhibit extra stability. $N = 2, 8, 20, 28, 50, 82, 126$ constitute the basic coordinates of the nuclear shell. However, independent particle

motion is not exact. Nucleons interact through complex many-body correlations—pairing, quadrupole deformation, tensor forces—effects that lie beyond the descriptive power of a static mean field. To remedy this deficiency, the shell-correction method (the Strutinsky method) was introduced as a phenomenological compensation for the discrepancy between independent particle motion and experiment (Strutinsky, 1967).

Meanwhile, the liquid drop model, which treats the nucleus as an incompressible charged droplet (Bethe & Bacher, 1936), has also achieved enormous success, reaching an $R^2 = 0.96$ in describing the macroscopic trend of binding energies. The core parameters of the liquid drop model—volume term, surface term, Coulomb term, symmetry energy term—derive from classical physical intuition and do not depend on the quantum-mechanical premise that nucleons move independently in a mean field. The shell model and the liquid drop model have long coexisted—one based on quantum-mechanical independent particle motion, the other on classical-physics macroscopic droplets—but the conceptual tension between them has never been systematically resolved.

This theoretical framework has achieved tremendous success. Yet beneath that success, a fundamental question has never been systematically interrogated: Is independent particle motion itself the fundamental organizing principle of the nuclear shell? Does the conceptual tension between the liquid drop model and the shell model point to a deeper unified structure? If both are only approximations, what is the true organizing principle?

1.2 The Factor Hierarchy Law and Testability Norms

The Factor Hierarchy Law provides a novel methodological framework for answering this question. The law states: In any complex system driven by multiple quantifiable factors, factors naturally divide into two hierarchical levels—regime factors and execution factors. Regime factors determine which set of rules currently applies; their function is manifested not in direct explanatory power but in the interaction effects through which they modulate the exposure coefficients of execution factors. When a regime factor crosses a critical value, the factor weights of the entire system are systematically reset (Tang, 2026g, 2026r).

The law comprises three axioms:

- **Axiom 1 (Factor Dichotomy):** In any complex system driven by multiple quantifiable factors, factors naturally divide into two hierarchical levels: regime factors and execution factors.
- **Axiom 2 (Regime Switch):** When a regime factor crosses a critical value, the factor weights of the entire system are systematically reset.
- **Axiom 3 (Interaction Effect):** The function of a regime factor is manifested not in direct explanatory power but in the interaction effects through which it modulates the exposure coefficients of execution factors.

The Factor Hierarchy Law is not a theoretical hypothesis—it is an empirical regularity that emerges automatically from objective data. Its discovery began in finance (Tang, 2026a-f) and subsequently obtained independent cross-disciplinary verification in astronomy (Tang, 2026h-j), statistical physics (Tang, 2026k-m),

particle physics (Tang, 2026n-q), and chemistry (Tang, 2026s-u). Nuclear physics is the sixth discipline to be deployed.

The operational implementation of the Factor Hierarchy Law is the Testability Norms—comprising six functional dimensions: search completeness, modulation relationship, structural breakpoints, causal direction, problem completeness, and process closure. The Testability Norms are not bound to any specific tools—the functional dimensions are eternal, and the implementation tools are variable (Tang, 2026g, 2026r).

1.3 Summary of Empirical Findings of the First Two Papers

Tang (2026v) deployed the Factor Hierarchy Law and Testability Norms to perform a zero-preset regime audit on the full AME2020 nuclide dataset, achieving four tiers of core findings: (1) the η hierarchy—continuous N/Z $\eta > 10^6$, binary magic-number marker $\eta = 19.19$, continuous shell-correction energy $\eta \approx 0$, establishing that under continuous-variable proxies N and Z behave as regime factors and the shell correction as an execution factor; (2) $N=14$ was confirmed as a statistically significant structural rupture through triple independent tool cross-validation; (3) the full- N -axis Bai-Perron sequential blind test locked onto an optimal 10 regime-switch points, with traditional magic numbers systematically shifted or skipped; (4) six quantitative theorems (Theorems 1–6) established the repositioning of traditional magic numbers within the statistical structure.

Tang (2026w) achieved five tiers of core findings on this foundation: (1) the Transition-Zone Width Theorem (Theorem 7) established the internal self-consistency of the confirmed-breakpoint network—1000 random breakpoint sets never reproduced 7 exact hits ($p < 0.001$); (2) shell-model coverage is only 15%, the odd-even effect jumps systematically at magic numbers, and the $N=Z$ line is identified as a regime boundary; (3) regime-switch signals disappear under the high-precision fitting of the liquid drop model; (4) compensation traces of the shell-correction term—three tools consistently avoid the shell-closure positions, cross-regime rupture ratio of mass covariance is 22.31; (5) the Cascade Skeleton Theorem (Theorem 8) established the exact cascade relationship among the confirmed breakpoints. Three new theorems (Theorems 7–9) and fifteen new discoveries together constitute a complete picture of the completeness audit of the shell model.

To facilitate independent reading, **Table 0a** summarizes the core conclusion data of Tang (2026v), and **Table 0b** provides a cross-paper theorem numbering reference.

Table 0a: Core Conclusion Data of Tang (2026v) (Inherited by This Paper)

Asset	Content
Regime Factors	N and Z ($\eta > 10^6$)
Execution Factors	$\text{mass_excess_keV} + \text{beta_decay_keV} + N_over_Z$ ($R^2 = 0.8052$)
Bai-Perron Optimal 10 Breakpoints	[15, 29, 65, 83, 104, 118, 129, 137, 145, 150]

Optimal BIC Value	13793.7
N=14 Precursor Signal	Confirmed by triple cross-validation (inv-family activation operators + threshold interaction + PICO)

Table 0b: Cross-Paper Theorem Numbering Reference

Source	Number	Name
Tang (2026v)	Theorems 1–6	Magic-Number Shift, Statistical Distinction, Magic-Number Node, N=14 Statistical Rupture, Regime-Factor Determination, Dimension Recovery
Tang (2026w)	Theorems 7–9	Transition-Zone Width Theorem (Nuclear-Physics Version), Confirmed-Breakpoint Cascade Skeleton Theorem, Switching-Type Spectrum Trend Theorem
This Paper	Theorems A–F	2D Ising Calibration Theorem, Nuclear-Shell Organizing Principle Theorem, Spherical-Symmetry Limit Recovery Theorem, Classical Limit Recovery Theorem, Independent-Particle-Motion Axiom Incompleteness Theorem, Triple Demotion Unification Theorem

1.4 Conceptual Distinction of the Triple Demotion and the Core Task of This Paper

The core task of this paper is to complete a unified argument for a triple demotion of the nuclear shell.

First demotion: the shell model. The shell model is not a fundamental theory—it is the projection of the regime-switch network in the spherical-symmetry limit. The core evidence is that the Chow F of the five traditional magic numbers is amplified by a factor of 3.68–9.68 under continuous sphericity weighting.

Second demotion: the liquid drop model. The liquid drop model is not an independent phenomenological model—it is the projection of the regime-switch network in the classical limit. The core evidence is the QCTI tending to -0.81 in the heavy-nucleus region, the strict monotonic relationship between the intercept/slope ratio and the mean-field error, and the uniformity of the liquid-drop masking effect.

Third demotion: the independent-particle-motion axiom. This is the most fundamental demotion. Independent particle motion—an axiom regarded for nearly eight decades as the fundamental organizing principle of the nuclear shell—is shown to be incomplete as a framework for the statistical description of many-body systems. It applies only within the regime interior under the quantum limit and cannot explain why magic numbers are not the optimal statistical partition points, why the shell correction requires such enormous external compensation, why shell-model coverage is only 15%, why the intercept/slope ratio increases monotonically with N , and why the QCTI decreases monotonically from $+0.94$ to -0.81 .

The shell model and the liquid drop model are each independently demoted—the shell model is the projection in the spherical-symmetry limit (Theorem C), and the liquid drop model is the projection in the classical limit (Theorem D). The fact that the two have not been unified for nearly nine decades is not because they share a mistaken premise, but because the quantum-classical hybrid framework reveals, for the first time, the limiting conditions under which each is applicable, providing a unified physical picture for their coexistence.

The argument structure of this paper is: 2D Ising calibration → seven-metric evidence matrix → triple demotion → unification.

It should be candidly acknowledged that all current lines of evidence derive from multi-dimensional re-analyses of the same set of AME2020 nuclear mass data, satisfying statistical independence but not data-source independence; independent verification across different data sources is a key direction for future work.

2. 2D Ising Exact Solution Calibration—Establishing the Physical Meaning of the Intercept/Slope Ratio

2.1 Full-Data Calibration with T_c as a Fixed Breakpoint

To establish the physical meaning of the intercept/slope ratio, we performed a calibration analysis on the Onsager exact solution of the 2D Ising model (Onsager, 1944; Yang, 1952). Taking the exact critical temperature $T_c = 2.269185$ as a fixed breakpoint, the full ferromagnetic-phase data ($T < T_c$, 295 data points) and the full paramagnetic-phase data ($T > T_c$, 205 data points) were separately fitted to an $M \sim T$ equation, yielding a baseline intercept/slope ratio of 10.81 for the 2D Ising model.

To test whether the intercept/slope ratio varies monotonically with the distance to T_c within the ferromagnetic phase, the ferromagnetic phase was divided into five temperature intervals—deep ferromagnetic region ($T < 1.0$), intermediate ferromagnetic region ($1.0 \leq T < 1.5$), mid-near- T_c region ($1.5 \leq T < 1.8$), near- T_c region ($1.8 \leq T < 2.1$), and T_c -neighborhood ($2.1 \leq T < 2.269$)—and each interval was paired with the full paramagnetic-phase data to compute the intercept/slope ratio. **Table 1** reports the intercept/slope ratios and their physical comparisons across the five temperature intervals.

Table 1: Intercept/Slope Ratio and Dual Physical-Quantity Comparison across Five Temperature Intervals of the 2D Ising Model

Temperature Interval	T Center	Distance to T_c	Intercept/Slope Ratio	$ dM/dT $	ϵ_{MF}
Deep Ferromagnetic ($T < 1.0$)	0.75	1.519	875.75	0.0015	0.0000
Intermediate Ferromagnetic ($1.0 \leq T < 1.5$)	1.25	1.019	41.98	0.0257	0.0013
Mid-Near- T_c ($1.5 \leq T < 1.8$)	1.65	0.619	11.62	0.0993	0.0096
Near- T_c ($1.8 \leq T < 2.1$)	1.95	0.319	5.19	0.2964	0.0437
T_c -Neighborhood ($2.1 \leq T < 2.269$)	2.18	0.085	2.67	2.5213	0.1814

The intercept/slope ratio across the five temperature intervals decreases monotonically from 875.75 in the deep ferromagnetic region to 2.67 in the T_c -neighborhood, Spearman $\rho = 1.0000$, $p = 0.0000$. The intercept/slope ratio in the deep ferromagnetic region is 81 times the baseline value. Based on this, we refine **Theorem A (2D Ising Calibration Theorem)**:

On the 2D Ising exact solution, the intercept/slope ratio $r(T)$ is a strictly monotonically increasing function of the distance to T_c , $d = |T - T_c|$ (Spearman $\rho = 1.0000$, $p = 0.0000$). The intercept/slope ratio takes a minimum value (2.67) in the fluctuation-dominated limit (near T_c) and a maximum value (875.75) in the mean-field-dominated limit (deep ferromagnetic), spanning a 328-fold range.

2.2 Dual Physical Verification—Dual Anchoring of Fluctuation Intensity and Mean-Field Description Failure

To further anchor the physical meaning of the intercept/slope ratio, a dual physical verification was performed within the 2D Ising model. The numerical derivative $|dM/dT|$ of the Onsager exact solution was used as a proxy for the susceptibility χ , and the root-mean-square error ϵ_{MF} between the mean-field approximation (self-consistent solution of $M_{MF} = \tanh(4M/T)$) and the exact solution was computed for each temperature interval.

The Spearman correlation between the intercept/slope ratio and $|dM/dT|$ is $\rho = -1.0000$ ($p = 0.0000$), and between the intercept/slope ratio and ϵ_{MF} is $\rho = -1.0000$ ($p = 0.0000$). Two independent physical curves—fluctuation intensity and mean-field description error—simultaneously exhibit strict negative correlations with the intercept/slope ratio. The closer to $T_c \rightarrow$ the smaller the intercept/slope ratio $\rightarrow$ the stronger the fluctuations $\rightarrow$ the worse the mean-field description. The farther from $T_c \rightarrow$ the larger the intercept/slope ratio $\rightarrow$ the weaker the fluctuations $\rightarrow$ the better the mean-field description.

In the 2D Ising model, the intercept/slope ratio measures the distance to the phase-transition critical point—the closer to T_c , the more the system behavior is dominated by fluctuations; the farther from T_c , the more the system behavior is dominated by the mean field. The 2D Ising calibration provides heuristic support in the form of a formal isomorphism for the "distance to critical point" interpretation of the intercept/slope ratio—this isomorphism is at the level of statistical regularity, not physical essence: the fluctuations in the 2D Ising model are thermal fluctuations, while those in nuclear physics are quantum fluctuations; the two are physically completely different but statistically isomorphic. The "quantum-classical transition" interpretation of the intercept/slope ratio in nuclear physics is independently anchored by the strong negative correlation with the shell-correction energy ΔR^2 (§3.4, $\rho = -0.87$)—the two lines of evidence are logically independent yet convergent in their conclusions. Theorem A itself is a mathematical fact internal to the 2D Ising model; $\rho = \pm 1.0000$ does not depend on any cross-system analogy.

2.3 Epistemological Status of the Calibration

The 2D Ising calibration provides three levels of positioning for the argument of this paper. First level: physical verification of tool sensitivity. On a system with known truth, the intercept/slope ratio indeed varies monotonically with distance ($\rho = \pm 1.0000$)—if it could not detect distance on a known system, it would be even

less likely to do so on an unknown system. This constitutes a necessary condition for the reliability of the tool. Second level: parallel evidence for the physical-metric pattern. In both systems, the same statistic covaries with physical metrics in the same pattern—the closer to the critical point, the stronger the fluctuations / the larger the quantum residual, and the smaller the intercept/slope ratio. This is a replication of a statistical regularity independent of the physical essence of the 2D Ising model. Third level: logical positioning of cross-system inference. The physical interpretation of the intercept/slope ratio in nuclear physics is empirical induction, not mathematical proof. The 2D Ising calibration establishes the physical meaning of this statistic on a known system; the strong negative correlation between the intercept/slope ratio and the shell-correction energy ΔR^2 in nuclear physics constitutes parallel evidence independent of the calibration—the two are in a relationship of parallel evidence, not causal derivation.

The validity of Theorem B does not depend on the 2D Ising calibration. The simplified QCTI analysis shows that after removing the intercept/slope ratio component, the QCTI still maintains strong monotonicity ($\rho = -0.93$), nearly identical to the $\rho = -0.95$ of the full QCTI. This means that even if the intercept/slope ratio component is completely removed, the core conclusion of the quantum-classical transition still holds. The 2D Ising calibration is an "added bonus" rather than a "necessary condition" for the argument—it provides physical intuition for the intercept/slope ratio calibrated by an exact solution, but the core empirical support for the quantum-classical hybrid framework comes from the seven-metric evidence matrix and the spherical-symmetry limit recovery within nuclear physics itself.

3. Multi-Dimensional Empirical Basis of the Quantum-Classical Hybrid

3.1 The QCTI Index and the Seven-Metric Evidence Matrix

To quantify the transition of the nuclear shell from quantum-effect-dominated to classical-effect-dominated, this paper constructs the Quantum-Classical Transition Index (QCTI), integrating three independent metrics: the shell-correction energy ΔR^2 , the binary marker ΔR^2 , and the intercept/slope ratio. The QCTI measures quantum residuals using the liquid drop model as the classical baseline; it is a "quantum residual measure relative to the liquid drop model" rather than a "model-independent measure of quantality." The Spearman correlation between the QCTI across 28 sliding windows on the full N-axis and N is $\rho = -0.9502$, $p = 0.0000$. The mean QCTI in the light-nucleus region ($N \leq 50$) is $+0.94$ (quantum-dominated), in the mid-shell region ($50 < N \leq 100$) is $+0.06$ (balanced), and in the heavy-nucleus region ($N > 100$) is -0.81 (classical-dominated).

To test the robustness of the QCTI, a simplified QCTI was constructed using only the shell-correction energy ΔR^2 and the binary marker ΔR^2 . The Spearman correlation between the simplified QCTI and N is $\rho = -0.9261$ ($p = 0.0000$), nearly identical to the $\rho = -0.9502$ of the full QCTI. The shell-correction energy ΔR^2

and the binary marker ΔR^2 are each independently and significantly correlated with N ($\rho = -0.8736$ and $\rho = -0.5677$, respectively, both $p < 0.01$). The core conclusion of the QCTI—the monotonic transition of the nuclear shell from quantum-dominated to classical-dominated—does not depend on the physical interpretation of the intercept/slope ratio, nor on the 2D Ising calibration.

The shell-correction energy ΔR^2 is extracted from the residuals of the liquid drop model, which already presupposes "the liquid drop model as the baseline." This means that there is an epistemological limitation to the physical interpretation of the QCTI: $\text{QCTI} \rightarrow 0$ does not necessarily mean "quantum effects have disappeared"; it could also mean that "the liquid drop model has already absorbed the quantum effects"—the current data cannot fully distinguish between these two interpretations. The strong negative correlation between the shell-correction energy ΔR^2 and the intercept/slope ratio (§3.4)—combined with the $\rho = -1.0000$ between the intercept/slope ratio and $|dM/dT|$ in the 2D Ising calibration—provides cross-validation that "the shell-correction energy ΔR^2 indeed captures quantum effects." The liquid drop model is the most successful classical baseline in nuclear physics ($R^2 = 0.96$); using it as a reference to measure quantum residuals is physically reasonable.

Table 2 summarizes the systematic zonal characteristics of the seven independent metrics across the three nuclear regions, constituting the seven-metric evidence matrix.

Table 2: Seven-Metric Evidence Matrix—Multi-Metric Zonal Characteristics of the Quantum-Classical Transition

Metric	Light-Nucleus Region ($N \leq 50$)	Mid-Shell Region ($50 < N \leq 100$)	Heavy-Nucleus Region ($N > 100$)	Trend
Intercept/Slope Ratio	21.7	70.5	129.4	Monotonically increasing
Liquid Drop Model R^2	0.7049	0.9420	0.9911	Monotonically increasing
Binary Marker Interaction ΔR^2	0.00145	0.000255	0.000011	132-fold decay
Shell-Correction Energy ΔR^2	0.0427	0.0278	0.0020	Inverse-U direction
$N_{\text{over}}Z$ Coefficient	−1466	−998	−60	24.4-fold jump
Activation-Operator Chow F	105.4	6.8	54.1	Mid-shell silence
Liquid-Drop/OLS Chow Ratio	0.26–0.36	0.28–0.36	0.29–1.46	Zonal proximity

The seven metrics come from five independent data dimensions and display systematic zonal characteristics across the three nuclear regions, collectively pointing in the same physical direction—the nuclear shell undergoes a smooth transition from quantum to classical.

3.2 Operational Definition of the Quantum-Classical Hybrid

The quantum-classical hybrid consists of three independently measurable components. The quantum-shell-effect component is measured by the shell-correction energy ΔR^2 , which takes a maximum value (~ 0.04) in the light-nucleus region and approaches zero (~ 0.002) in the heavy-nucleus region, physically corresponding to the degree of validity of independent particle motion. The classical-liquid-drop component is measured by the liquid drop model R^2 , which increases monotonically from 0.70 in the light-nucleus region to 0.99 in the heavy-nucleus region, physically corresponding to the degree of validity of macroscopic liquid-drop behavior. The regime-switch component is measured by the intercept/slope ratio $r(N)$, which increases monotonically from ~ 22 in the light-nucleus region to ~ 129 in the heavy-nucleus region, physically corresponding to the distance to the phase-transition critical point (Theorem A), and modulates the relative weights of the first two components.

Bivariate statistical characterization: The quantum-classical hybrid is statistically described by two independent coordinates—the shell-correction energy ΔR^2 (quantum axis) and the intercept/slope ratio $r(N)$ (classical-quantum transition axis). The strong negative correlation between the two coordinates ($\rho = -0.87$) indicates that they capture different facets of the same transition process. Univariate description: The spectrum value $S(N)$ —the intercept/slope ratio linearly mapped to the interval $[0, 1]$ —serves as a single measure of the "quantum-classical mixing ratio." $S \rightarrow 1$ corresponds to quantum-effect dominance, and $S \rightarrow 0$ corresponds to classical-effect dominance.

The domain of validity of the quantum-classical hybrid framework is: $S > 0.7$ (light-nucleus region), the shell model is a good approximation, and the liquid drop model provides only the macroscopic background; $0.3 < S < 0.7$ (mid-shell region), neither limit is fully applicable, and the degree of quantum-classical mixing is highest; $S < 0.3$ (heavy-nucleus region), the liquid drop model is a good approximation, and the shell correction approaches zero.

In this paper, the "quantum-classical hybrid" is a statistical description framework, and its three components—the shell-correction energy ΔR^2 , the liquid drop model R^2 , and the intercept/slope ratio—are all statistical quantities. The core claim of this framework is that the statistical structure of the nuclear shell can be unifiedly described by the quantum-classical mixing ratio, and that this description framework is systematically superior to the independent-particle-motion framework in terms of internal self-consistency and descriptive completeness. It does not contain the dynamical mechanisms of quantum mechanics or classical physics, but rather provides a statistical characterization of the relative importance of these mechanisms in different nuclear regions.

3.3 The Double-Track Intersection Point $N \approx 70.8$

The intercept/slope ratio increases monotonically from light to heavy nuclei, while the shell-correction energy ΔR^2 first increases and then decreases from light to heavy nuclei. The two curves intersect exactly at

$N \approx 70.8$, which lies in the mid-shell region, approximately 13 neutrons from the mid-shell center at 84. $N \approx 70.8$ is the dividing line between quantum-effect dominance and classical-effect dominance—in the region $N < 70.8$, quantum effects are statistically more significant; in the region $N > 70.8$, classical effects are statistically more significant. The RMSE in the $N = 68\text{--}74$ interval around the intersection point decreases monotonically, confirming that the intersection point is not the "optimal prediction partition point" but rather the "quantum-classical equilibrium point."

This intersection point emerges automatically from two completely independent statistics—the shell-correction energy ΔR^2 comes from the η -hierarchy structure of Tang (2026v), and the intercept/slope ratio comes from Theorem 9 of Tang (2026w)—and is entirely unrelated to the construction process of the Bai-Perron blind-detected breakpoint network. Yet the Bai-Perron optimal breakpoints $N=65$ (before the intersection) and $N=83$ (after the intersection) happen to flank the intersection point on both sides.

To further verify the uniqueness of the double-track intersection, the seven metrics in Table 2 were paired pairwise, generating 21 pairs of intersection analyses. For each pair of metrics, the intersection points of the two curves were searched after normalization over the range $N = 20\text{--}150$. The intersection point between the shell-correction energy ΔR^2 and the intercept/slope ratio is the only metric pair with a clear and stable intersection characteristic: the two cross from opposite directions, the crossing angle is large, and they do not repeatedly cross after the intersection. The liquid drop model R^2 and the intercept/slope ratio increase monotonically in the same direction and only "approach" rather than "cross" at $N \approx 50\text{--}60$. The intersection point between the binary marker ΔR^2 and the intercept/slope ratio appears at $N \approx 30\text{--}40$ (light-to-medium nucleus transition region). The activation-operator Chow F collapses into a near-zero narrow band in the mid-shell region and crosses multiple metrics multiple times—this is a "noise zone" rather than an "equilibrium point." The seven-metric intersection matrix confirms: the intersection point between the shell-correction energy ΔR^2 and the intercept/slope ratio is the only point among the 21 pairs with a clear and stable intersection characteristic, and this uniqueness rules out the alternative explanation that "the double-track intersection is merely an accidental intersection of two arbitrary metrics."

3.4 Strong Negative Correlation between Shell-Correction Energy ΔR^2 and Intercept/Slope Ratio

The Spearman correlation between the shell-correction energy ΔR^2 and the intercept/slope ratio is $\rho = -0.8707$, $p = 0.0000$. This strong negative correlation directly opens a quantitative channel between "distance to the critical point" and the quantum residual—the closer to the critical point, the larger the quantum residual; the farther from the critical point, the smaller the quantum residual. This correlation provides empirical anchoring within nuclear physics itself for the "quantum-classical transition" interpretation of the intercept/slope ratio,

independent of the 2D Ising calibration. Based on the above empirical foundation, we refine **Theorem B (Nuclear-Shell Organizing Principle Theorem)**:

The nuclear shell is a quantum-classical hybrid system; N and Z , as semi-continuous regime factors, modulate the regime switch between quantum shell effects and classical liquid-drop behavior. The shell model is an effective approximation in the quantum limit, and the liquid drop model is an effective approximation in the classical limit. The intercept/slope ratio $r(N)$ is interpreted in nuclear physics as a statistical proxy for the quantum-classical mixing ratio: this interpretation is cross-validated by two independent lines of evidence—the 2D Ising calibration provides heuristic support in the form of a formal isomorphism (at the level of statistical regularity), and the strong negative correlation with the shell-correction energy ΔR^2 provides empirical anchoring within nuclear physics itself (at the level of physical correspondence). The two lines of evidence are logically independent yet convergent in their conclusions. The QCTI measures quantum residuals using the liquid drop model as the classical baseline—the liquid drop model is the most successful classical baseline in nuclear physics ($R^2 = 0.96$); using it as a reference is physically reasonable.

3.5 The Mid-Shell Silence Zone—The Region of Most Intense Quantum-Classical Competition

The activation-operator zonal-sensitivity analysis reveals an important structure: the mid-shell region is the common silence zone of all activation operators. The activation-operator Chow F drops to 6.8 in the mid-shell region, only 1/15 of the light-nucleus region's 105.4. Bootstrap resampling with 10,000 iterations confirms that the 95% confidence interval of the mid-shell-region Chow F is [5.9, 7.7], completely below the light-nucleus-region mean of 105.4 and the heavy-nucleus-region mean of 54.1. The silence determination of the six activation operators is completely consistent—Kendall's coefficient of concordance W confirms that all operators, without exception, rank the mid-shell region as having the lowest Chow F .

The mid-shell silence is not an accidental performance of a particular operator but an inherent characteristic of the data structure. The binding-energy variance in the mid-shell region is only 1/20.8 of that in the light-nucleus region (41,104 vs. 853,250), and the variance of the rate of change of binding energy along the N -axis is only 1/22.3 of that in the light-nucleus region. The core sensitivity of the Chow test depends on the variance heterogeneity of the data between two regression segments—the low variance in the mid-shell region directly leads to the systematic suppression of the Chow F . It should be candidly acknowledged that the mid-shell silence is a superposition of a physical phenomenon and a statistical-test-power issue: the variance of the binding energy along the N -axis is indeed lowest in the interval $50 < N \leq 100$ (a physical fact), but whether this is the sole manifestation of quantum-classical competition, or whether there exist other unrecognized structures, requires more independent detection methods to distinguish.

The mid-shell silence zone and the liquid-drop masking zone are significantly negatively correlated in space—the Spearman correlation between the activation-operator Chow F and the liquid-drop/OLS Chow ratio is $\rho = -0.6661$ ($p = 0.0001$). Both are based on the Chow test framework, but the former measures the structural rupture intensity of the original data, while the latter measures the proportion of signal absorbed by the liquid drop model—they are mathematically related but physically independent. The mid-shell region is both the place where activation operators fail and the place where the liquid drop model absorbs signals most thoroughly—two independent statistical phenomena converge in the same physical region.

3.6 Divergence between Binary-Marker Decay and Shell-Correction Peak

The local interaction effect ΔR^2 of the binary magic-number marker decays significantly and monotonically across 28 sliding windows on the full N-axis, Spearman $\rho = -0.57$, $p = 0.0016$, from a mean of 0.00145 in the light-nucleus region to 0.000011 in the heavy-nucleus region—a 132-fold decay. This is direct evidence for the systematic disappearance of quantum shell effects: the binary magic-number marker captures discrete quantum shell effects, and this effect tends to disappear completely in the heavy-nucleus region.

The full-N-axis distribution of the shell-correction energy ΔR^2 exhibits a different trend: a mean of 0.0427 in the light-nucleus region, 0.0278 in the mid-shell region, and 0.0020 in the heavy-nucleus region. The maximum values of the shell-correction energy ΔR^2 appear near $N=50$ and $N=55$ (0.0703 and 0.0601, respectively), exactly at the starting point of the mid-shell region of the quantum-classical transition. A quadratic regression test shows that the inverse-U trend holds directionally (the N^2 coefficient is negative) but does not reach statistical significance ($p = 0.33$).

The opposing trends of "one decay, one peak"—quantum shell effects decaying while the demand for shell correction peaks in the mid-shell region—constitute the most central statistical fingerprint of the quantum-classical hybrid. The shell correction is not a simple function of quantum effects but a function of the degree of quantum-classical mixing: in the region of most intense mixing (the mid-shell region), quantum effects are no longer sufficient to independently describe the system, but classical effects have not yet fully dominated, so the demand for the shell correction as a quantum residual reaches its maximum.

3.7 Correspondence between Activation-Operator Stratification and Quantum-Classical Mixing Ratio

Tang (2026v) discovered that activation operators automatically stratify into two groups—all three inv-family operators lock onto $N=14$, and all four raw-family operators lock onto $N=30-32$. The correspondence between this stratification and the quantum-classical hybrid framework is as follows: the inv transformation amplifies the variance differences in the light-nucleus region (high quantum-effect region, extremely small intercept/slope ratio), making the inv-family operators highly sensitive to the light-nucleus regime switch at

N=14; the raw transformation preserves the global variance structure, making the raw-family operators produce a maximal response at the center of gravity of the quantum-classical transition zone at N=30–32.

The 2D Ising calibration provides a mathematical foundation for this interpretation: on the 2D Ising exact solution, different mathematical transformations alter the alignment between the explained variable and the sensitivity structure of the Chow test— χ is completely silent under the Chow test ($F = 16.8$), but $1/\chi$ activates it to a magnitude of $\sim 10^{21}$ (Tang, 2026k). The stratification behavior in nuclear physics—the inv family locking onto N=14 and the raw family locking onto N=30–32—follows the same mathematical logic: the differences in the quantum-classical mixing ratio across different N intervals are the physical basis on which different activation operators selectively illuminate.

3.8 Spectrum-Value Self-Consistency Check

Taking the minimum (N=14, 12.49) and maximum (N=129, 164.17) of the intercept/slope ratio within nuclear physics as anchor points, the intercept/slope ratio at each breakpoint is linearly mapped to the interval [0, 1] to construct the quantum-classical spectrum value S. The Spearman correlation between the spectrum value and the shell-correction ΔR^2 is $\rho = -0.8165$, $p = 0.0000$ —a strong negative correlation: the closer to the classical limit, the smaller the quantum residual. The correlation between the spectrum value and the liquid drop model R^2 is $\rho = 0.0653$ —directionally correct (positive correlation) but not significant, mainly because the liquid drop model R^2 has already reached a ceiling of 0.99 in the heavy-nucleus region, leaving almost no room for variation as the spectrum value continues to increase. The spectrum value at the double-track intersection point $N \approx 70.8$ is 0.407, close to the theoretically predicted 0.5—at the quantum-classical equilibrium point, the spectrum value is indeed near the midpoint. The three self-consistency checks jointly confirm the physical validity of the quantum-classical spectrum.

4. First Demotion: The Shell Model—Spherical-Symmetry Limit Recovery

4.1 Systematic Integration of Five Lines of Evidence and the Criterion Framework

The empirical findings of Tang (2026v) and Tang (2026w) can be integrated into five lines of evidence, which from four different dimensions jointly point in the same direction: independent particle motion is insufficient to serve as the fundamental organizing principle of the nuclear shell. **Table 3** summarizes the five lines of evidence.

Table 3: Systematic Integration of Five Lines of Evidence

Evidence	Core Content	Source	Argument Strength	Dimension
Evidence	Interaction effects of N and Z far exceed	Tang	High	Statistical

Evidence	Core Content	Source	Argument Strength	Dimension
1	direct effects ($\eta > 10^6$)	(2026v)		
Evidence 2	Bai-Perron blind test skips N=20 and N=50	Tang (2026v)	High	Statistical
Evidence 3	Shell-model coverage only 15–20% (directional observation)	Tang (2026w)	Directional Observation	Completeness
Evidence 4	Liquid-drop masking effect: $R^2=0.96$ but residual signals disappear	Tang (2026w)	Medium-High	Descriptive
Evidence 5	Shell-correction compensation traces: three tools consistently avoid	Tang (2026w)	Medium	Self-Consistency

The uniformity of liquid-drop masking holds under three window widths ($\pm 8, \pm 12, \pm 16$). The mean liquid-drop/OLS Chow ratios across the three nuclear regions are close under all three window widths, and the MWU test p-values for light vs. heavy nuclei are all non-significant (0.50, 0.60, 0.41), supporting the "absorption" hypothesis—the liquid drop model absorbs signals in approximately the same manner at any regime switch; the masking is uniform rather than selective.

The criterion framework consists of three core criteria and one auxiliary criterion. **Core Criterion 1 (Presupposition-Free Discovery Capability and Assumption Economy)**: The regime-switch network requires no mean-field assumption; it is more economical in assumptions and can discover structures that presupposition-dependent frameworks cannot. **Core Criterion 2 (Descriptive Completeness)**: The regime-switch network identifies 20 blind-detected breakpoints; the shell model can explain only a small fraction of them (directional observation). **Core Criterion 3 (Descriptive Integrity)**: The regime-switch network covers new structures that the shell model cannot. **Auxiliary Criterion (Internal Self-Consistency)**: The shell model requires the shell-correction term to remedy its deficiencies; the regime-switch network requires no external compensation.

The three core criteria logically share the premise of "shell-model incompleteness"—they are not three independent argumentative paths but three dimensional unfoldings of the same premise. This premise (shell-model incompleteness) has been jointly established by the five lines of evidence from four independent dimensions.

The incompleteness of the shell model is "organizing-principle incompleteness" rather than "domain-of-validity incompleteness." The incompleteness of Newtonian gravity is domain-of-validity incompleteness—it fails in strong-field regimes but is self-consistent in weak-field regimes. The incompleteness of the shell model is organizing-principle incompleteness—it not only requires the shell-correction term to remedy quantitative discrepancies but also cannot derive the main structural characteristics of the regime-switch network: Why are the Bai-Perron optimal breakpoints at N=29 and N=83 rather than N=28 and N=82? Why does the confirmed-

breakpoint network satisfy the Transition-Zone Width Theorem? Why do N=20 and N=28 share N=23 as a cascade node? Why does the intercept/slope ratio increase monotonically with N? Why does the QCTI decrease monotonically from +0.94 to -0.81?

4.2 The Methodological-Priority Theorem of Nuclear-Shell Regime Switching

Based on the above lines of evidence and criteria, we formally propose the **Methodological-Priority Theorem of Nuclear-Shell Regime Switching**:

"The traditional shell model takes 'nucleons move independently in a static mean field' as its axiom—magic numbers are its corollary, and the shell correction is its supplement. The Factor Hierarchy Law reveals that this axiom is insufficient to serve as the fundamental organizing principle of the nuclear shell. A more effective organizing principle of the nuclear shell is the many-body correlation regime switch modulated by N and Z as regime factors. Independent particle motion is not the fundamental organizing principle—it is the approximate manifestation of the regime switch in the spherical-symmetry limit. The shell correction is not a supplement to independent particle motion—it is a compensation for the discrepancy between the independent-particle-motion approximation and the true regime-switch structure. The shell model is the projection of the regime-switch network under the independent-particle approximation, not a first principle."

Formalized statement:

$$\begin{aligned} \text{Regime_switch}(N, Z) &\rightarrow \text{Multi_body_correlation}(\text{mode}) \rightarrow \text{Shell_structure} \\ \text{Shell_model} &= \text{Projection}(\text{Regime_switch_network}, \text{spherical_symmetry_limit}) \end{aligned}$$

"Methodological priority" is strictly defined as "more effective in discovering unknown structures," and does not imply physical-causal priority or epistemological priority. Methodological priority does not deny the quantitative precision of the shell model—the organizing principle answers "what is a more effective description framework," while quantitative precision answers "what is a more precise computational tool."

4.3 Magic-Number Recovery in the Spherical-Symmetry Limit—Core Evidence for the First Demotion

A continuous sphericity weight function is constructed based on the sum of distances to the nearest magic numbers: $w(N, Z) \propto \exp(-dN^2/2\sigma^2) + \exp(-dZ^2/2\sigma^2)$, where $dN = \min|N - N_{\text{magic}}|$, $dZ = \min|Z - Z_{\text{magic}}|$. The choice $\sigma = 3.0$ is based on the following independent physical reasoning: nuclides near traditional magic numbers are usually spherical or near-spherical within $\pm 2-3$ neutrons—the width of a magic-number shell closure is about 2–3 neutrons. A half-width at half-maximum of $\sigma/2 \approx 1.5$ neutrons means that the sphericity weight drops to about 61% of its peak at about 3 neutrons from a magic number. Sensitivity tests with $\sigma = 2.0-5.0$ confirm that the qualitative conclusion is robust to the choice of σ .

It should be candidly acknowledged that using the distance to a magic number as a proxy for sphericity draws on an empirical output of the shell model—the regularity that nuclides near magic numbers are spherical has been independently verified by numerous experiments (e.g., multiple independent measurements such as transition probabilities, nuclear moments, and isotope shifts), and this experimental fact does not depend on whether we believe that independent particle motion is the fundamental organizing principle. However, the validity of the magic-number distance as a proxy for sphericity depends in part on the physical logic of the shell model for "why nuclides near magic numbers are spherical." The WLS test utilizes the empirically confirmed components of the shell model's empirical output; its argument strength is positioned as "statistically strictly significant."

Table 4 reports the results of the WLS test ($\sigma = 3.0$).

Table 4: WLS Magic-Number Chow F Amplification Factors ($\sigma = 3.0$)

Magic Number N	OLS Chow F	WLS Chow F	Amplification Factor
20	1231.59	4532.09	3.68
28	1119.63	7739.28	6.91
50	834.80	5726.28	6.86
82	627.89	4293.41	6.84
126	392.60	3800.26	9.68

Under the four values of $\sigma = 2.0$ – 5.0 , the amplification factors of the five traditional magic numbers are all greater than 1.5 (minimum amplification 2.68 at $\sigma = 2.0$, minimum amplification 1.87 at $\sigma = 5.0$), and the results have been successfully reproduced in an independent environment. $\sigma = 3.0$ is a reasonable but not unique baseline choice; the complete argument depends on the robustness that all amplifications exceed 1.5 for $\sigma = 2.0$ – 5.0 . The amplification factors differ across magic numbers (N=126 amplified by a factor of 9.68 vs. N=20 amplified by a factor of 3.68)—this difference is one of the theoretical predictions: the wider the shell closure, the weaker the signal under OLS, and the greater the amplification under WLS.

To further verify the degradation mechanism of the shell model in the spherical-symmetry limit, the monotonic variation of the WLS amplification factor with σ was examined. As σ increases—the sphericity weight changing from a narrow Gaussian to a broad Gaussian—the WLS amplification factor should gradually decrease. When $\sigma \rightarrow \infty$, all weights approach a constant, and WLS degenerates back to OLS. The empirical data confirm this monotonic decreasing trend: the amplification factor is highest at $\sigma = 2.0$, and drops to its lowest at $\sigma = 5.0$ but remains > 1.5 . The continuous sphericity weight indeed captures the continuous variation of sphericity—it is not a binary "spherical/non-spherical" distinction, but "continuous variation of sphericity $\rightarrow$ continuous variation of the degree of approximation of the shell model."

Based on this, we refine **Theorem C (Spherical-Symmetry Limit Recovery Theorem)**:

Under continuous sphericity weighting (based on the sum of distances to the nearest magic numbers, $\sigma = 3.0$), the Chow F of the five traditional magic numbers is amplified by a factor of 3.68–9.68. In the spherical-symmetry limit, the status of traditional magic numbers as the optimal statistical partition points is recovered. The qualitative conclusion is robust for $\sigma = 2.0$ –5.0, and has been successfully reproduced in an independent environment. The shell model is the projection of the regime-switch network in the spherical-symmetry limit.

4.4 Physical Explanation of Shell-Model Degradation in the Spherical-Symmetry Limit

In the spherical-symmetry limit—when a nuclide is exactly at a traditional magic number, the sphericity weight takes its maximum value—the regime-switch effect of many-body correlations is suppressed by the dominant role of quantum-mechanical single-particle levels. At this point, the exposure coefficient of N_over_Z in the binding-energy equation drops to a minimum, and the binding energy is mainly determined by the mass excess and the decay energy—two variables that happen to be equivalent to the sum of single-particle energies in the shell model. The core axiom of the shell model, "nucleons move independently in a static mean field," is the exact degradation of the quantum-classical hybrid in the spherical-symmetry limit; the mathematical condition for the degradation is that the sphericity weight approaches 1. The shell model has not been falsified—it has been derived under specific conditions.

The above exposition is a qualitative physical explanation rather than a rigorous mathematical proof. Rigorously deriving the equations of the shell model from the regime-switch network would require starting from the first-principles many-body Schrödinger equation, which is beyond the scope of this paper. However, the physical explanation is sufficient to elucidate the physical root of the effectiveness of the shell model in the spherical-symmetry limit.

4.5 Theoretical Self-Consistency between Magic-Number Recovery and Magic-Number Shift

The conclusion of Theorems 1–3 (Tang, 2026v) is: magic numbers are not the optimal statistical partition points—this is under the conditions of the full N-axis, no weighting, and no prior in OLS. The conclusion of Theorem C is: under continuous sphericity weighting, the magic-number signals are systematically recovered—this is under weighted, conditional, restricted conditions. The two are not contradictory—they describe the different statistical performances of magic numbers under different physical conditions.

The operation of the continuous sphericity weight $w(N,Z)$ is to "give higher weight to spherical nuclides"—it does not assume that magic numbers are valid, but rather assumes that "spherical nuclides are more likely to exhibit independent-particle-motion characteristics," an assumption that does not depend on the status of magic numbers as the optimal statistical partition points. The Chow F amplification under WLS is evidence that magic numbers do approach the optimal statistical partition points in the spherical-symmetry

limit—Theorems 1–3 point out that "magic numbers are not," and Theorem C points out that "they are recovered as such in a specific limit." Together, the two establish a complete picture of magic numbers in the statistical structure: on the full nuclide chart, magic numbers are geometric nodes (Theorem 3); in the spherical-symmetry limit, magic numbers are recovered as statistical centers of gravity (Theorem C).

5. Second Demotion: The Liquid Drop Model—Classical Limit Recovery

5.1 QCTI Tending to the Classical Limit in the Heavy-Nucleus Region

The mean QCTI in the heavy-nucleus region is -0.81 (negative value = classical-effect dominated). The QCTI decreases monotonically with N ($\rho = -0.95$)—classical effects systematically strengthen in the heavy-nucleus region.

5.2 Strict Monotonic Relationship between Intercept/Slope Ratio and Mean-Field Error

2D Ising calibration: intercept/slope ratio vs. ε_{MF} : $\rho = -1.0000$. The larger the intercept/slope ratio $\rightarrow$ the more accurate the mean-field approximation $\rightarrow$ the better the classical description. Nuclear physics correspondence: the intercept/slope ratio in the heavy-nucleus region is extremely large (mean 129.4) $\rightarrow$ classical liquid-drop behavior dominates.

5.3 Positive Direction of Correlation between Spectrum Value and Liquid Drop Model R^2

The positive direction holds (ceiling effect limits significance). The liquid drop model R^2 increases monotonically from 0.70 in light nuclei to 0.99 in heavy nuclei—the classical description becomes increasingly accurate.

5.4 Verification of the "Absorption" Hypothesis of the Liquid-Drop Masking Effect

In the heavy-nucleus region, the signal-to-noise ratio of the liquid-drop residuals is higher than that of the OLS residuals (by a factor of 5.62), but the Chow F is systematically lower (by a factor of 0.31). The liquid-drop/OLS Chow ratio does not differ significantly across nuclear regions—the masking is uniform. The "absorption" hypothesis holds: the liquid drop model encodes regime-switch signals into its fitting parameters.

5.5 Theorem D (Classical Limit Recovery Theorem)

Theorem D (Classical Limit Recovery Theorem): The liquid drop model is the projection of the regime-switch network in the classical limit. In the classical limit, the intercept/slope ratio takes a maximal value, the shell-correction energy ΔR^2 approaches zero, and the liquid drop model R^2 approaches 1.

6. Third Demotion: The Independent-Particle-Motion Axiom—The Most Fundamental Demotion

6.1 The Independent-Particle-Motion Axiom—A Premise Never Questioned

The axiom of the shell model is that "nucleons move independently in a static mean field"—magic numbers are its corollary, and the shell correction is its supplement. This axiom has never been systematically questioned. The unique feature of the triple demotion is that it demotes not only the shell model itself but also the axiom on which the shell model depends—the validity of independent particle motion as a framework for the statistical description of many-body systems.

The liquid drop model does not depend on the independent-particle-motion axiom—its core parameters come from classical physical intuition. The fact that the shell model and the liquid drop model have not been unified for nearly nine decades is not because they share a mistaken premise, but because the quantum-classical hybrid framework reveals, for the first time, the limiting conditions under which each is applicable: the shell model is the exact degradation of the quantum-classical hybrid in the spherical-symmetry limit (Theorem C), and the liquid drop model is the exact degradation of the quantum-classical hybrid in the classical limit (Theorem D). The two are each independently demoted, and under the quantum-classical hybrid framework, they obtain a unified physical picture for the first time.

6.2 Conceptual Distinction between Quantum-Mechanical Precision and Statistical-Description Sufficiency

Before discussing the incompleteness of the axiom, two independent conceptual levels must be distinguished. Quantum mechanics is precise in predicting the positions of magic numbers—the magic numbers $N = 2, 8, 20, 28, 50, 82, 126$, as quantum-mechanical nodes of single-particle levels, are determined by the Schrödinger equation plus spin-orbit coupling. All the findings of this paper—Theorems 1–3 (magic-number shift), Theorem 4 ($N=14$ statistical rupture), Theorem 7 (Transition-Zone Width Theorem)—do not question the precision of these magic numbers as quantum-mechanical nodes. The blind-detected breakpoints of Tang (2026v, 2026w) are highly close to these magic numbers (deviations of 1–15 neutrons), and this proximity is itself independent evidence for the precision of quantum mechanics.

However, independent particle motion is insufficient as a framework for the statistical description of many-body systems. The shell correction requires such enormous external compensation ($\eta > 10^6$, shell-correction energy ΔR^2 reaching 0.07 in the mid-shell region), the Bai-Perron blind test skips $N=20$ and $N=50$, and shell-model coverage is only 15%—these are not failures of the quantum-mechanical nodes themselves, but failures of independent particle motion as a statistical-description framework. Quantum-mechanical precision and statistical-description sufficiency are two different properties: the former only requires finding the shell

closures at the correct positions; the latter requires describing all nuclides with the same equation. The real error of the independent-particle-motion axiom is that it slid from the precision of the former into the sufficiency of the latter—it was taken to hold where it does not hold.

6.3 Lines of Evidence and Quantitative Evidence for the Incompleteness of the Axiom

Regularities that independent particle motion cannot explain include: Theorems 1–3 (magic-number shift), Theorem 4 (N=14 statistical rupture), Theorem 7 (Transition-Zone Width Theorem), Theorem 8 (Cascade Skeleton Theorem), Theorem 9 (monotonic increase of the intercept/slope ratio), monotonic decrease of the QCTI ($\rho = -0.95$), the mid-shell silence zone, and the double-track intersection point.

Individual ΔBIC values of magic numbers: $\Delta\text{BIC} = -514.4$ for N=20 (forcing N=20 into the breakpoint set severely deteriorates the fit), $\Delta\text{BIC} = +415.9$ for N=28 (overwhelming evidence), $\Delta\text{BIC} = +11.9$ for N=82 (marginal improvement, only about 1/35 of that for N=28). If magic numbers are the fundamental organizing principle, why does forcing N=20 as a breakpoint into the model severely deteriorate the fit? Why is the improvement for N=82 only about 1/35 of that for N=28? Shell-model coverage is only 15%. The η hierarchy: the continuous shell-correction energy is an execution factor ($\Delta R^2 = 0.343$), not a regime factor.

6.4 Theorem E (Independent-Particle-Motion Axiom Incompleteness Theorem)

Theorem E (Independent-Particle-Motion Axiom Incompleteness Theorem):

Theorem E-1 (At the Level of Organizing Principle): Independent particle motion is not the fundamental organizing principle of the nuclear shell. A more effective organizing principle of the nuclear shell is the quantum-classical hybrid—N and Z, as semi-continuous regime factors, modulate the regime switch between quantum shell effects and classical liquid-drop behavior. The evidence comes from the convergence of five independent lines of evidence: Evidence 1 ($\eta > 10^6$, many-body correlations dominate), Evidence 2 (magic numbers are not the optimal partition points), Theorem 7 (self-consistency of the confirmed-breakpoint network), Theorem 8 (cascade skeleton), and Theorem 9 (monotonic increase of the intercept/slope ratio).

Theorem E-2 (At the Level of Domain of Validity): Independent particle motion is an effective approximation of the quantum-classical hybrid in the quantum limit. It applies within the regime interior under the spherical-symmetry limit, within which the shell-correction term compensates for the discrepancy between it and the true structure. The evidence comes from Theorem C (WLS all-five-magic-number amplification) and Evidence 5 (shell-correction compensation traces).

Theorem E-1 answers "whether independent particle motion is the fundamental organizing principle"—it is not. Theorem E-2 answers "under what conditions independent particle motion is a useful approximation"—within the regime interior under the spherical-symmetry limit. The two levels jointly establish the repositioning of independent particle motion: not fundamental, but useful. What Theorem E denies is the sufficiency of

independent particle motion as a framework for the statistical description of many-body systems—not the validity of quantum mechanics in nuclear physics.

7. Unification of the Triple Demotion and Completion of Derivational Demotion

7.1 Logical Hierarchy of the Triple Demotion

The shell model and the liquid drop model are each independently demoted. The shell model is demoted because it is the projection in the spherical-symmetry limit (Theorem C)—with no direct dependence on the independent-particle-motion axiom. The liquid drop model is demoted because it is the projection in the classical limit (Theorem D)—with no direct dependence on the independent-particle-motion axiom. The independent-particle-motion axiom is demoted (Theorem E) because it is shown to be incomplete as a framework for the statistical description of many-body systems—shell-model coverage is only 15%, and the shell correction requires external compensation. The triple demotion has a clear source-to-stream relationship in logical hierarchy: Theorem C and Theorem D establish the respective limiting conditions of the two models, Theorem E establishes the incompleteness of the axiom, and Theorem F unifies all three into the quantum-classical hybrid framework.

7.2 Theorem F (Triple Demotion Unification Theorem)

Theorem F (Triple Demotion Unification Theorem): The shell model, the liquid drop model, and the independent-particle-motion axiom are each projections of the quantum-classical hybrid in specific limits. The shell model is the projection in the spherical-symmetry limit, the liquid drop model is the projection in the classical limit, and the independent-particle-motion axiom is an approximation in the quantum limit. The statistical organizing principle of the nuclear shell is the quantum-classical hybrid— N and Z , as semi-continuous regime factors, modulate the regime switch between quantum shell effects and classical liquid-drop behavior.

7.3 Assessment of the Completion of Derivational Demotion

The completion of derivational demotion must simultaneously satisfy two necessary conditions: (a) the new theory is superior to the old theory in discovering unknown structures—corresponding to Conditions 5–8 (WLS all-five-magic-number amplification, semi-continuous confirmed-breakpoint spacing, Chow F zonal regime, positive direction of correlation between spectrum value and liquid-drop R^2); (b) the core predictions of the old theory can be degraded from the new theory in a specific limit—corresponding to Conditions 1–4 (2D Ising calibration establishes the physical meaning of the intercept/slope ratio, monotonic increase of the intercept/slope ratio confirms the quantum-classical transition, strong negative correlation between spectrum

value and shell-correction ΔR^2 opens the quantitative channel, double-track intersection point locates the quantum-classical equilibrium point). Condition 2 and Condition 3 share the core statistic of the intercept/slope ratio and are logically partially related. The simplified QCTI analysis shows that after removing the intercept/slope ratio, the QCTI still maintains strong monotonicity ($\rho = -0.93$); therefore, although Condition 2 and Condition 3 share the same core statistic, they carry information about different facets in the argument—Condition 2 confirms the monotonicity of the quantum-classical transition itself, while Condition 3 confirms the quantitative relationship between this transition and the quantum residual.

Table 5 reports the assessment of the completion of derivational demotion. All 8 conditions are satisfied.

Table 5: Assessment of the Completion of Derivational Demotion

Condition	Status	Independence Annotation
1. 2D Ising calibration $\rho = 1.0000$ + dual physical verification $\rho = -1.0000$	✓	Independent
2. Monotonic increase of intercept/slope ratio $\rho > 0.9$	✓	Partially related to Condition 3
3. Strong negative correlation between spectrum value and shell-correction ΔR^2 $ \rho > 0.7$	✓	Partially related to Condition 2
4. Spectrum value at double-track intersection point close to 0.5	✓	Independent
5. WLS all-five-magic-number amplification > 1.5 (robust for $\sigma = 2.0\text{--}5.0$)	✓	Independent
6. CV of confirmed-breakpoint spacing > 0.5 (semi-continuous)	✓	Independent
7. Chow F zonal CV < 1 (zonal regime)	✓	Independent
8. Positive direction of correlation between spectrum value and liquid-drop R^2	✓	Independent

7.4 Completion of Derivational Demotion in the Three Hard-Science Disciplines

Table 6 systematically compares the demotion arguments of the three disciplines across ten dimensions.

Table 6: Comparison of the Demotion Arguments of the Three Hard-Science Disciplines

Comparison Dimension	Particle Physics	Chemistry	Nuclear Physics
Demoted theory	Gauge field theory	Periodic law	Shell model, liquid drop model
Traditional axiom	Gauge symmetry	Group-period classification	Independent particle motion
Regime-switch corresponding	Color charge $\rightarrow$ SU(3)	period/valence $\rightarrow$ s,p,d,f	N/Z $\rightarrow$ blind-detected breakpoint network

structure			
Regime-factor type	Purely discrete	Purely discrete	Semi-continuous
Positioning after demotion	Regime switch expressed in the continuous-symmetry limit	Regime switch projected in group-period coordinates	Regime switch projected under the independent-particle approximation
Core theorem	Gauge Group Structure Theorem	Chemical Gauge Group Theorem	Methodological-Priority Theorem + Triple Demotion Theorem
Repositioning of compensation term	—	—	Shell-correction term = quantum residual
Demotion type	Constructive derivation	Constructive derivation	Limit-degradation derivation
Argument strength	High (mathematically rigorous proof)	High (group-theoretically rigorous derivation)	High (statistically rigorous verification)
Source	Tang (2026p, 2026q)	Tang (2026u)	This paper

The derivational demotion in particle physics and chemistry is a constructive derivation of the form "if A, then B": given the three color-charge states $\rightarrow$ SU(3); given the eight valence-electron states $\rightarrow$ s, p, d, f. The derivational demotion in nuclear physics is a limit-degradation derivation of the form "if A, then in limit L, A degrades to B": given N and Z as regime factors, in the spherical-symmetry limit, it degrades to the shell model; in the classical limit, it degrades to the liquid drop model. All three are derivational demotions, but the form of "derivation" differs—constructive derivation is direct derivation, and limit-degradation derivation is conditional derivation; both are mathematically rigorous. The limit-degradation derivation in nuclear physics is no less rigorous than the constructive derivation—the spherical-symmetry limit degradation has been rigorously verified by the WLS all-five-magic-number amplification, and the classical limit degradation has been rigorously anchored by the QCTI monotonicity and the dual physical verification.

8. Testable Theoretical Predictions

Based on the triple-demotion framework and the multi-dimensional empirical basis of the quantum-classical hybrid presented in this paper, seven testable theoretical predictions are proposed. Each prediction is accompanied by a specific verification path and a falsification condition.

Prediction 1 (Internal to Nuclear Physics): The QCTI in the superheavy-nucleus region ($N > 150$) continues to decrease or tends to saturate. The quantum-classical hybrid framework predicts that quantum effects tend to vanish in the classical limit. If the applicability of the liquid drop model in the superheavy-

nucleus region significantly declines ($R^2 < 0.9$), then the monotonic decreasing trend of the QCTI may partly reflect the failure of the liquid drop model baseline rather than the disappearance of quantum effects. Verification path: superheavy-nucleus mass measurement data + QCTI calculation. Falsification condition: if the QCTI rebounds in the superheavy-nucleus region (> 0).

Prediction 2 (Internal to Nuclear Physics): The shell-correction energy ΔR^2 in the superheavy-nucleus region further approaches zero. Verification path: superheavy-nucleus mass measurement data + shell-correction energy extraction. Falsification condition: if the shell-correction energy ΔR^2 does not decrease but instead increases in the superheavy-nucleus region.

Prediction 3 (Internal to Nuclear Physics): Bai-Perron should identify more secondary breakpoints in the $N = 65\text{--}83$ interval. The double-track intersection point $N \approx 70.8$ lies between the Bai-Perron optimal breakpoints $N=65$ and $N=83$, which is the core region of the quantum-classical transition. Verification path: higher-precision nuclear mass data + Bai-Perron blind test. Falsification condition: if no new breakpoints are identified in the $N = 65\text{--}83$ interval with higher-precision data.

Prediction 4 (Internal to Nuclear Physics): The candidate breakpoints $N=25$ and $N=32$ in the light-nucleus region will be confirmed as genuine regime-switch points by future high-precision experiments. Verification path: high-precision binding-energy measurements in the light-nucleus region + Chow test. Falsification condition: if no significant Chow signal appears.

Prediction 5 (Cross-Disciplinary): Mid-shell-type silence zones exist in quantum-classical hybrid systems in condensed matter physics. The quantum-classical hybrid framework predicts that any quantum-classical hybrid system should exhibit systematic failure of a single detector in the region where the two effects are equally matched. Verification path: deploy multi-operator Chow tests on conductance-fluctuation data from semiconductor quantum-dot arrays and superfluid-insulator transition data from cold-atom optical lattices. Falsification condition: if no silence zone is found in three or more quantum-classical hybrid systems.

Prediction 6 (Cross-Disciplinary): Double-track intersection points exist in molecular excited states in quantum chemistry. Verification path: calculate the non-adiabatic coupling strength (quantum-effect metric) and the nuclear kinetic energy (classical-effect metric) on the potential energy surfaces of molecular excited states, and search for their intersection point. Falsification condition: if the two metrics have no intersection point.

Prediction 7 (Cross-Disciplinary): The r-process waiting points ($N = 137, 145, 150$) in nuclear astrophysics should exhibit regime-switch signals on more precise mass data. Verification path: more precise nuclear mass measurements on the r-process path + Bai-Perron blind test. Falsification condition: if Bai-Perron detects no significant breakpoints (BIC improvement < 10).

Table 7 summarizes the verification paths and falsification conditions of the seven predictions.

Table 7: Seven Testable Predictions and Their Falsification Conditions

Prediction	Domain	Verification Path	Falsification Condition
QCTI trend in superheavy region	Nuclear physics	Superheavy-nucleus data + QCTI	QCTI rebounds > 0
Shell-correction decay in superheavy region	Nuclear physics	Superheavy-nucleus data + shell-correction extraction	ΔR^2 does not decrease but increases
Bai-Perron secondary breakpoints	Nuclear physics	High-precision nuclear masses + Bai-Perron	No new breakpoints
Confirmation of light-nucleus candidate breakpoints	Nuclear physics	High-precision binding energies + Chow	No significant signal
Condensed-matter-physics silence zone	Condensed matter physics	Multi-operator Chow test	No silence zone
Quantum-chemistry double-track intersection point	Quantum chemistry	Non-adiabatic coupling calculation	No intersection point
r-process waiting-point signal	Nuclear astrophysics	Precision masses + Bai-Perron	No significant breakpoint

9. Discussion and Conclusion

9.1 Resolution of Nearly Nine Decades of Conceptual Tension

The shell model and the liquid drop model are no longer two independent fundamental theories—they are projections of the quantum-classical hybrid in two limits. Independent particle motion is no longer the fundamental organizing principle—the quantum-classical hybrid has replaced it. The empirical basis for this unification includes the 2D Ising exact-solution calibration (Theorem A), the seven-metric evidence matrix (Table 2), spherical-symmetry limit recovery (Theorem C), classical limit recovery (Theorem D), and axiom incompleteness (Theorem E); five independent paths converge to the same conclusion.

9.2 Theoretical Significance of the Triple Demotion

First demotion (shell model): projection in the spherical-symmetry limit. Second demotion (liquid drop model): projection in the classical limit. Third demotion (independent-particle-motion axiom): the most fundamental demotion—the axiom itself is downgraded. The triple demotion shares the same empirical foundation and is unified by Theorem F.

9.3 Paradigm Shift in Nuclear Physics Theory

From "independent particle motion" to "quantum-classical hybrid"—nuclear physics theory has undergone a fundamental transformation of its basic assumptions. **Table 8** systematically compares the two paradigms across nine dimensions.

Table 8: Nine-Dimensional Comparison of the Paradigm Shift

Comparison Dimension	Independent-Particle-Motion Paradigm	Quantum-Classical Hybrid Paradigm
Core axiom	Nucleons move independently in a static mean field	Regime switch of the quantum-classical hybrid modulated by N and Z
Magic-number positioning	Fundamental constants	Geometric nodes of shell-closure transition zones
Shell-correction function	"Supplement" to independent particle motion	Quantum residual—compensating for quantum-classical differences
Liquid-drop model positioning	Independent phenomenological model	Effective approximation in the classical limit
Regime-factor type	—	N and Z: semi-continuous regime factors
Organizing principle	Independent particle motion	Regime switch of the quantum-classical hybrid
Descriptive completeness	Coverage only 15%	Coverage 100% (20 blind-detected breakpoints)
Domain of validity	Regime interior under the spherical-symmetry limit	Full nuclide chart (N = 14–150)
Theoretical unity	Shell model and liquid drop model coexist independently	Unified as two limits of the quantum-classical hybrid

The theoretical framework of the new paradigm has been established, and its logical self-consistency and empirical support have been confirmed through multi-tool cross-validation and independent reproduction. Whether the paradigm shift is accepted by the scientific community depends on the verification of the predictions of this paper by future experiments, reproduction by independent teams, and the actual guiding effect of the new paradigm on nuclear physics research.

9.4 Candid Disclosure of Argument-Strength Grading

The argument of this paper can be graded into four levels of strength. **Rigorous mathematical proof:** Theorem A ($\rho = \pm 1.0000$). **Statistically strictly significant:** Theorem B ($\rho = -0.95$), Theorem C (WLS all-five-magic-number amplification, independently reproducible), Theorem E (five lines of evidence converging from four independent dimensions). **Directional support:** Liquid-drop model demotion (positive direction of correlation between spectrum value and liquid-drop R^2 , significance limited by ceiling effect), doubly-magic-

nucleus deformation proxy ($\rho = 0.87$, $p = 0.054$, marginally significant). **Exploratory observation:** Small-sample limitations of triple screening, divergence between binary-marker decay and shell-correction peak, physical explanation of shell-model degradation mechanism (§4.4).

All current lines of evidence derive from multi-dimensional re-analyses of the same set of AME2020 nuclear mass data, satisfying statistical independence but not data-source independence. Theorem C has been successfully reproduced in an independent environment (§4.3); cross-data-source verification of the remaining theorems requires independent experimental data.

9.5 Conclusion

This paper has completed the triple demotion of the nuclear shell—the shell model, the liquid drop model, and the independent-particle-motion axiom have each been shown to be projections of the quantum-classical hybrid in specific limits. Six theorems (Theorems A–F) have been refined, and all 8 conditions for the completion of derivational demotion have been satisfied. The statistical organizing principle of the nuclear shell is the quantum-classical hybrid— N and Z , as semi-continuous regime factors, modulate the regime switch between quantum shell effects and classical liquid-drop behavior.

This paper has completed the derivational demotion in nuclear physics, which, together with the demotion of gauge field theory in particle physics and the demotion of the periodic law in chemistry, constitutes the complete demotion argument of the Factor Hierarchy Law in the three hard-science disciplines.

Data Availability Statement

This paper is a theoretical-derivation and system-integration paper. The empirical foundations of all axioms and premises have been reported in detail in Tang (2026v) and Tang (2026w). The supplementary analyses of this paper (2D Ising calibration, dual physical verification, QCTI construction, mid-shell silence-zone identification, double-track intersection-point location, WLS test, and independent-reproduction verification) are based on re-analyses of the empirical data already published in the first two papers. The data are completely public, and any researcher can independently reproduce all results.

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